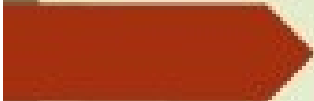


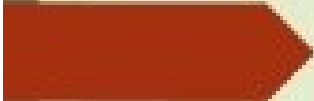
Multiple Access

FDMA, TDMA, CDMA, OFDMA and SDMA



BACKGROUND

- Multiple Access techniques are used to allow many mobile users to share simultaneously a finite amount of radio spectrum.
- The sharing of spectrum is required to achieve high capacity by simultaneously allocating the available bandwidth (or the available amount of channels) to multiple users.
- There are five basic access or multiple access methods: frequency division multiple access (FDMA), time division multiple access (TDMA), code division multiple access (CDMA), orthogonal frequency division multiple access (OFDMA), and spatial division multiple access (SDMA).



DIFFERENCE BETWEEN MULTIPLE ACCESS AND MULTIPLEXING

MULTIPLE ACCESS

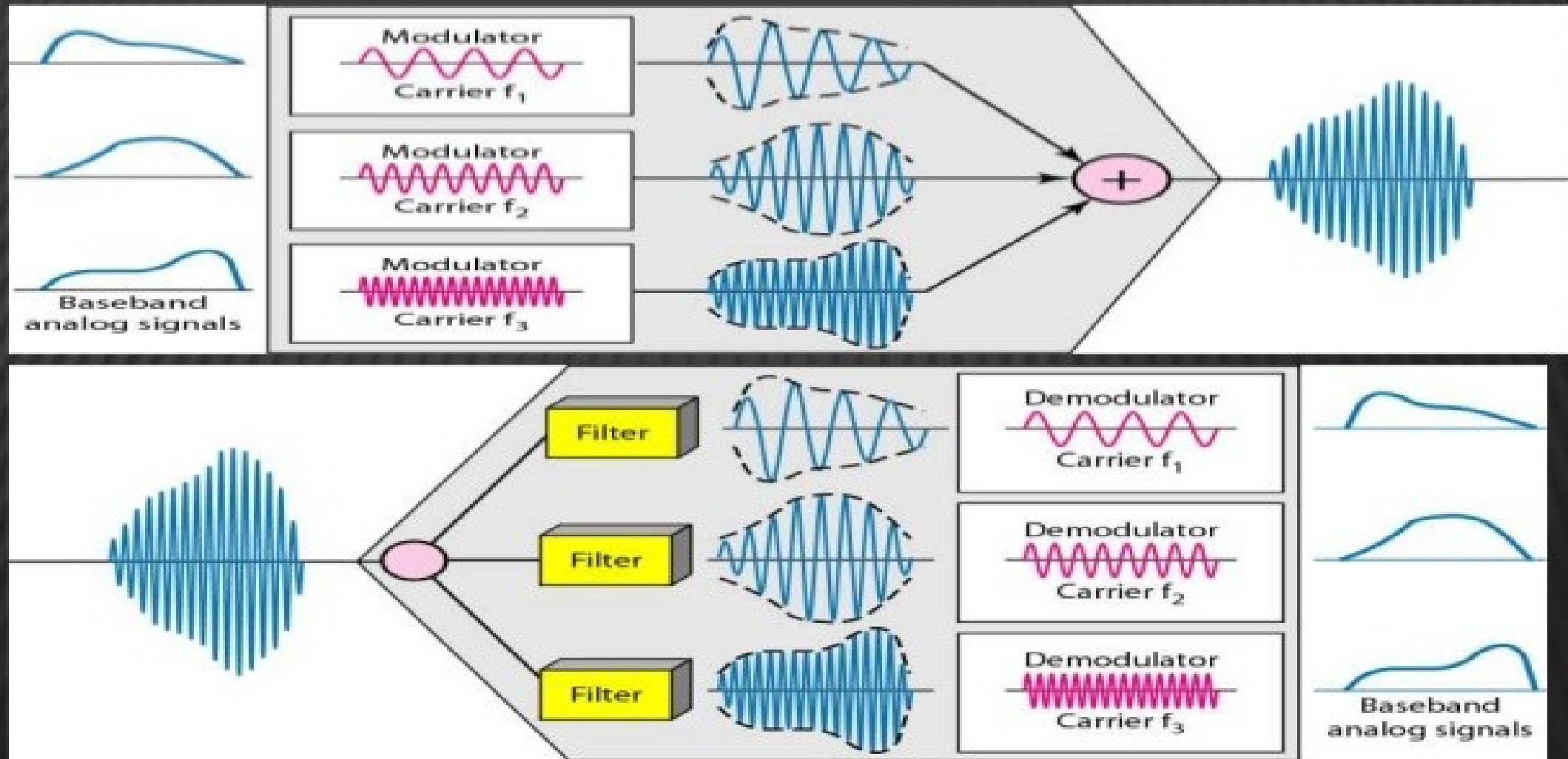
It is a technique that permit many users to simultaneously access a given frequency allocation.

MULTIPLEXING

It is process to combine multiple signals for transmit it over a single channel or media.

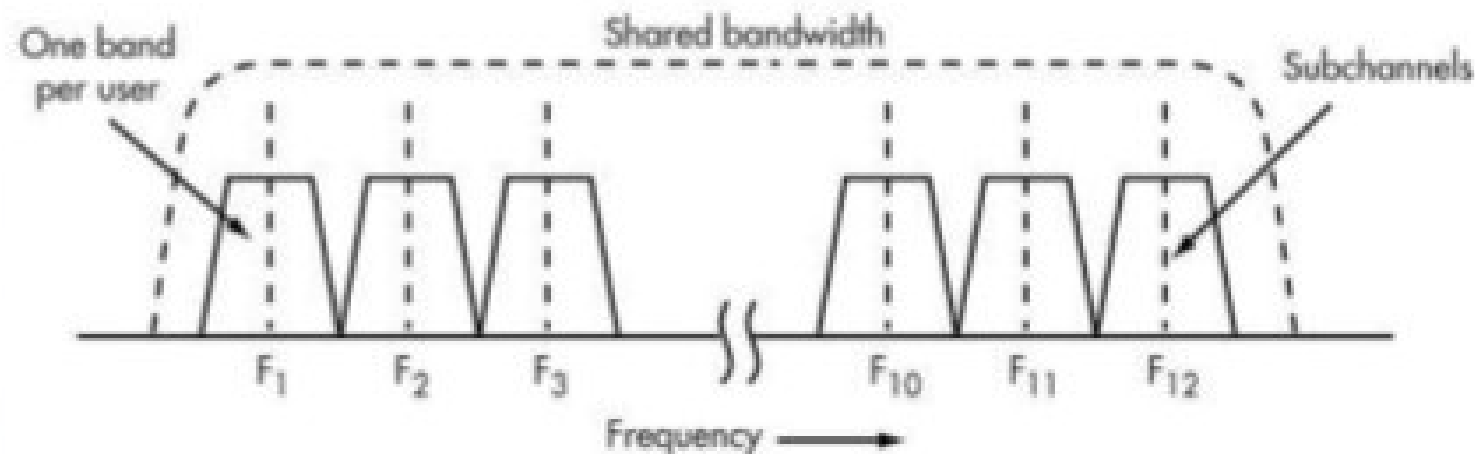
FREQUENCY
DIVISION
MULTIPLE
ACCESS (FDMA)

Frequency division multiplexing

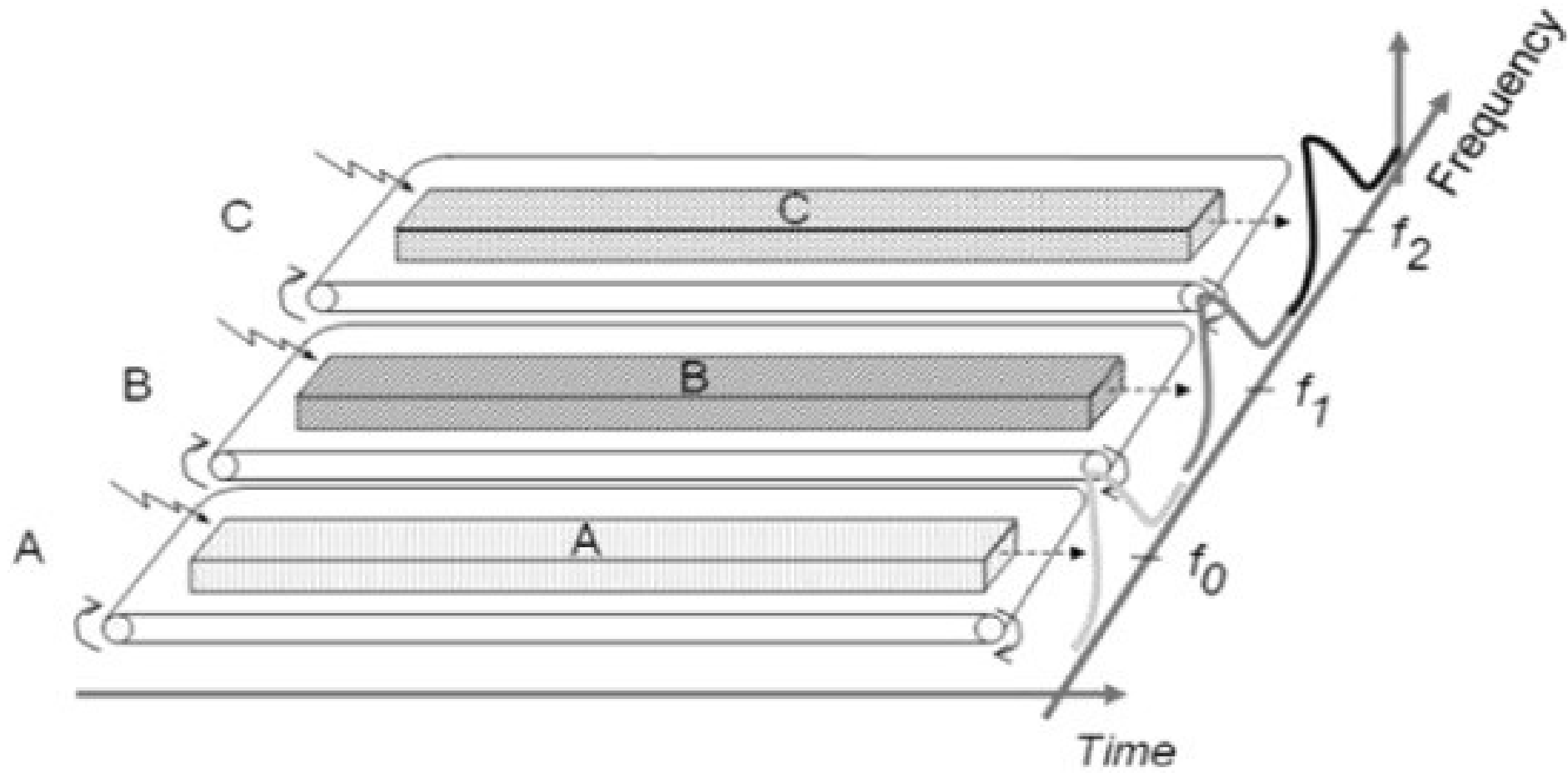


INTRODUCTION

- FDMA is the division of the frequency bands allocated for wireless cellular telephone communication into channels.
- Each channel can be assigned to only one user at a time
- All users transmit and receive at different frequencies because every user receives an individual frequency slot.



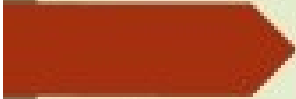
FDMA





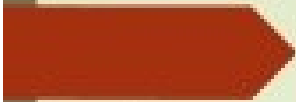
HOW IT WORKS?

- FDMA allows multiple users to send data through a single communication channel, such as a coaxial cable or microwave beam, by dividing the bandwidth of the channel into separate non overlapping frequency sub channels.
- Then allocating each sub channel to a separate user.
- Users can send data through a sub channel by modulating it on a carrier wave at the sub channel's frequency.



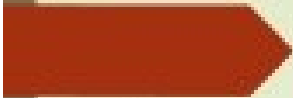
CONT..

In FDMA method, guard bands are used between the adjacent frequency slots to minimize crosstalk between the channels. A specific frequency band is given to one person, and it will be received by identifying each of the frequency on the receiving end. It is often used in the first generation of analog mobile phone.



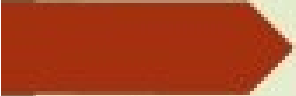
USES OF FDMA:

- FDMA is used in the Total Access Communication System (TACS).
- It is used in satellite communication systems and telephone lines.
- The Digital-Advanced Mobile Phone Service (D-AMPS) also uses FDMA but adds time division multiple access(TDMA)



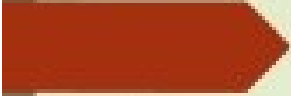
ADVANTAGES OF FDMA

- Due to the frequency filtering, FDMA is not sensitive to near- far problem.
- Many of the problems experienced in TDMA due to different propagation delays are eliminated in FDMA as there is no need for network timing
- Fairly efficient when the number of stations is small and the traffic is uniformly constant
- No restriction regarding the type of baseband or type of modulation



DISADVANTAGES OF FDMA

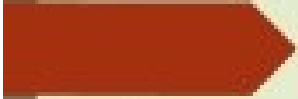
- One disadvantage of FDMA is crosstalk, which can cause interference between frequencies and interrupt the transmission.
- Between the different used frequency channels is a small amount of bandwidth not used.
- The maximum flow rate per channel is fixed and small.
- Relatively expensive and complicated band pass filters required



EXAMPLES

CABLE TELEVISION SYSTEM

- The best example of this is the cable television system. The medium is a single coax cable that is used to broadcast hundreds of channels of video/audio programming to homes.
- The coax cable has a useful bandwidth from about 4 MHz to 1 GHz. This bandwidth is divided up into 6-MHz wide channels.



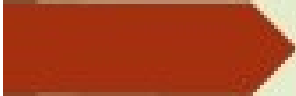
EXAMPLES

ANALOG TELEPHONE SYSTEM

- One of the older FDMA systems is the original analog telephone system, which used a hierarchy of frequency multiplex techniques to put multiple telephone calls on single line.

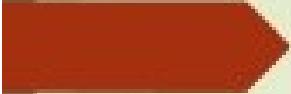


TIME DIVISION MULTIPLE ACCESS (TDMA)



WHAT IS TDMA TECHNOLOGY?

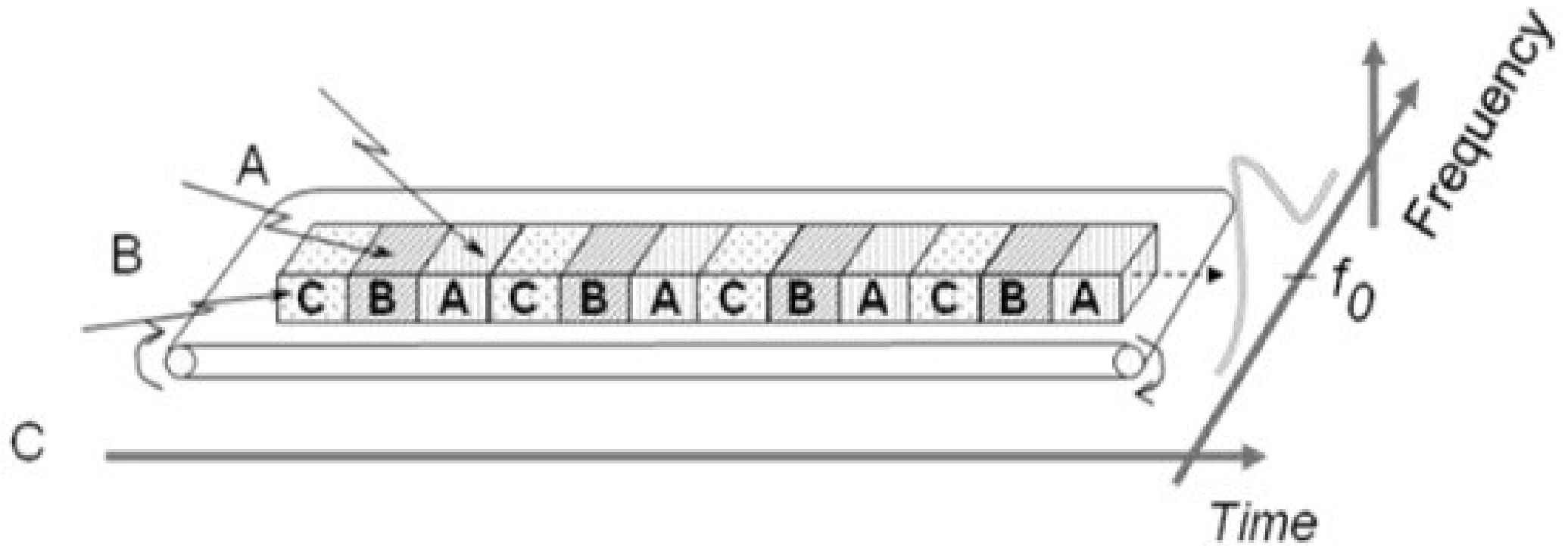
- Time Division Multiple Access(TDMA) is a digital cellular telephone communication technology.
- It facilitates many users to share the same frequency without interference.
- It's technology divides a signal into different time slots.
- Hence increase the data carrying capacity.



HOW IT WORKS?

- TDMA takes a cellular communication channel(frequency band) and slices it in to a series of time segments.
- Each cellular user is assigned the time slices with a given number.
- Each user transmit information only for the duration of their time segments using TDMA scheme.
- By Small time segments and High slicing frequency user Perceives a continuous Communication channel.

TDMA



FDMA VS TDMA

FDMA

Entire band of frequencies is divided into multiple RF channels/carriers. Each carrier allocated to different users.

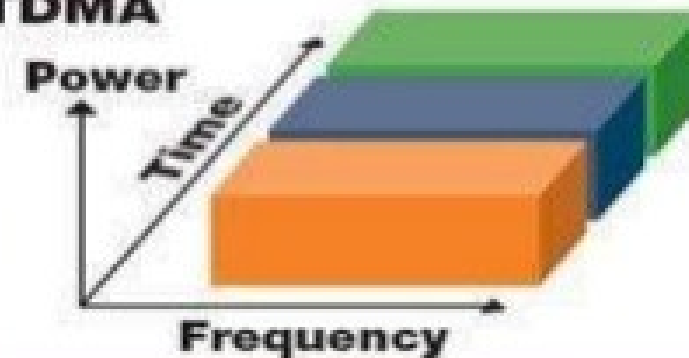
FDMA

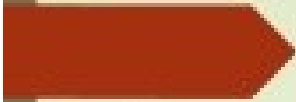


TDMA

It's entire bandwidth is shared among different subscribers at fixed predetermined or dynamically assigned time intervals.

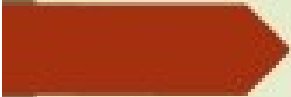
TDMA





ADVANTAGES

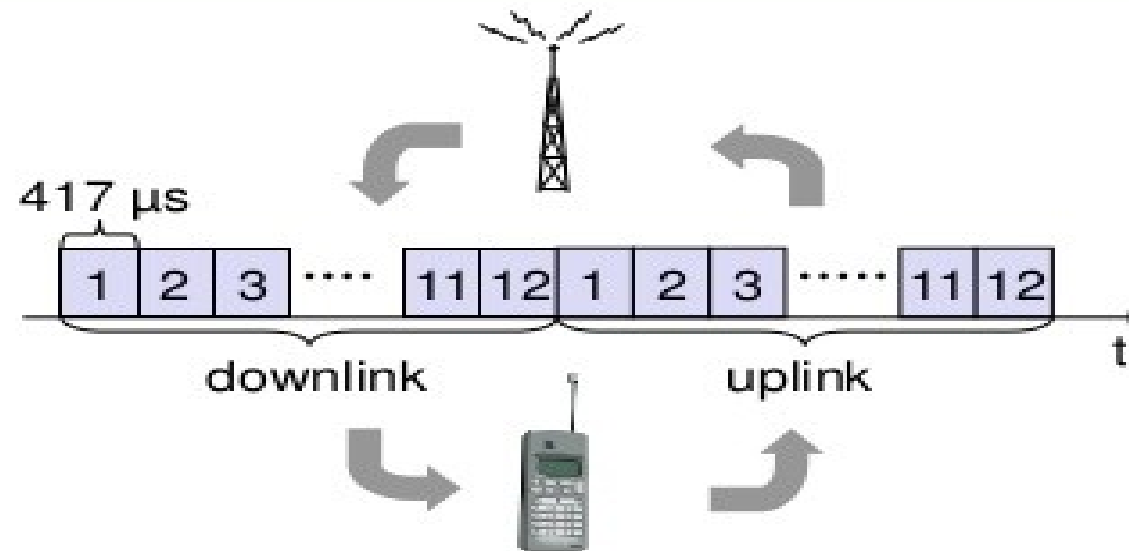
- Carry data rates from 64 kbps to 120 Mbps
- More number of users can use same spectrum.
- Operational costs are lower than traditional FDMA.
- Different types of traffics easily achieved by allocation of different time slots.
- Resources utilized effectively as per need basis and battery life enhanced.



DISADVANTAGES

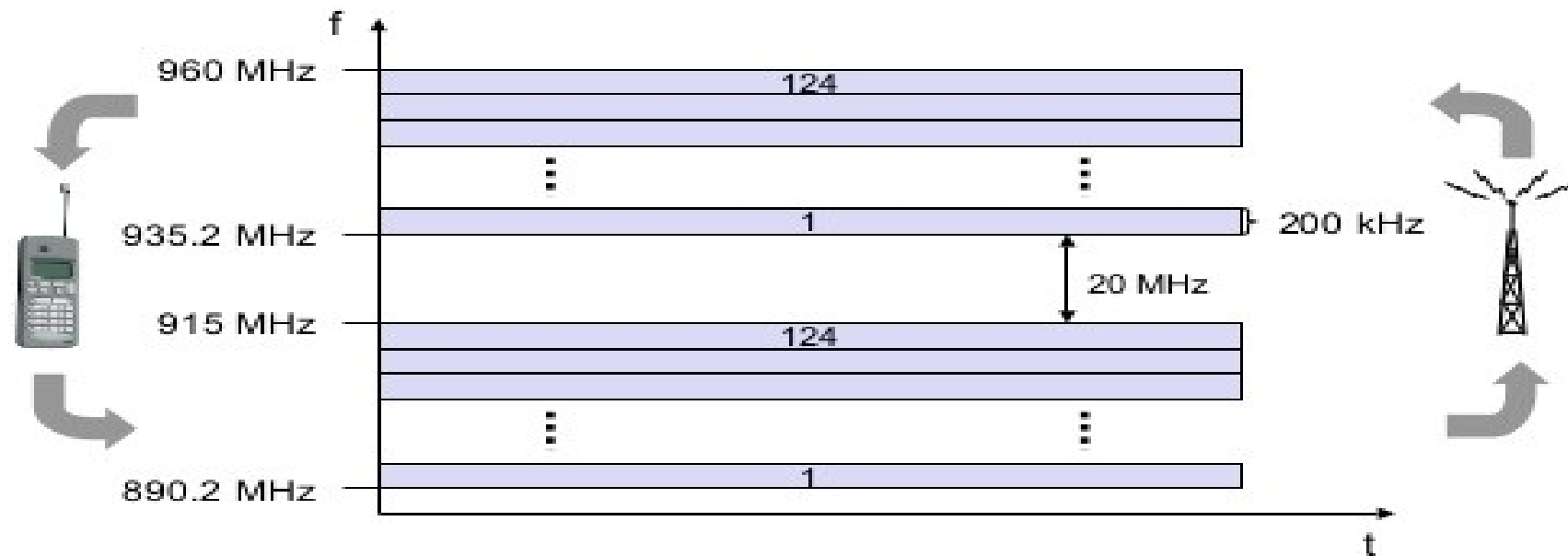
- ▶ Network and spectrum planning require more efforts.
- ▶ Multipath interference affects call quality.
- ▶ Switching from one base station cell to the other cell results into dropped calls.

TDMA/TDD – example: DECT

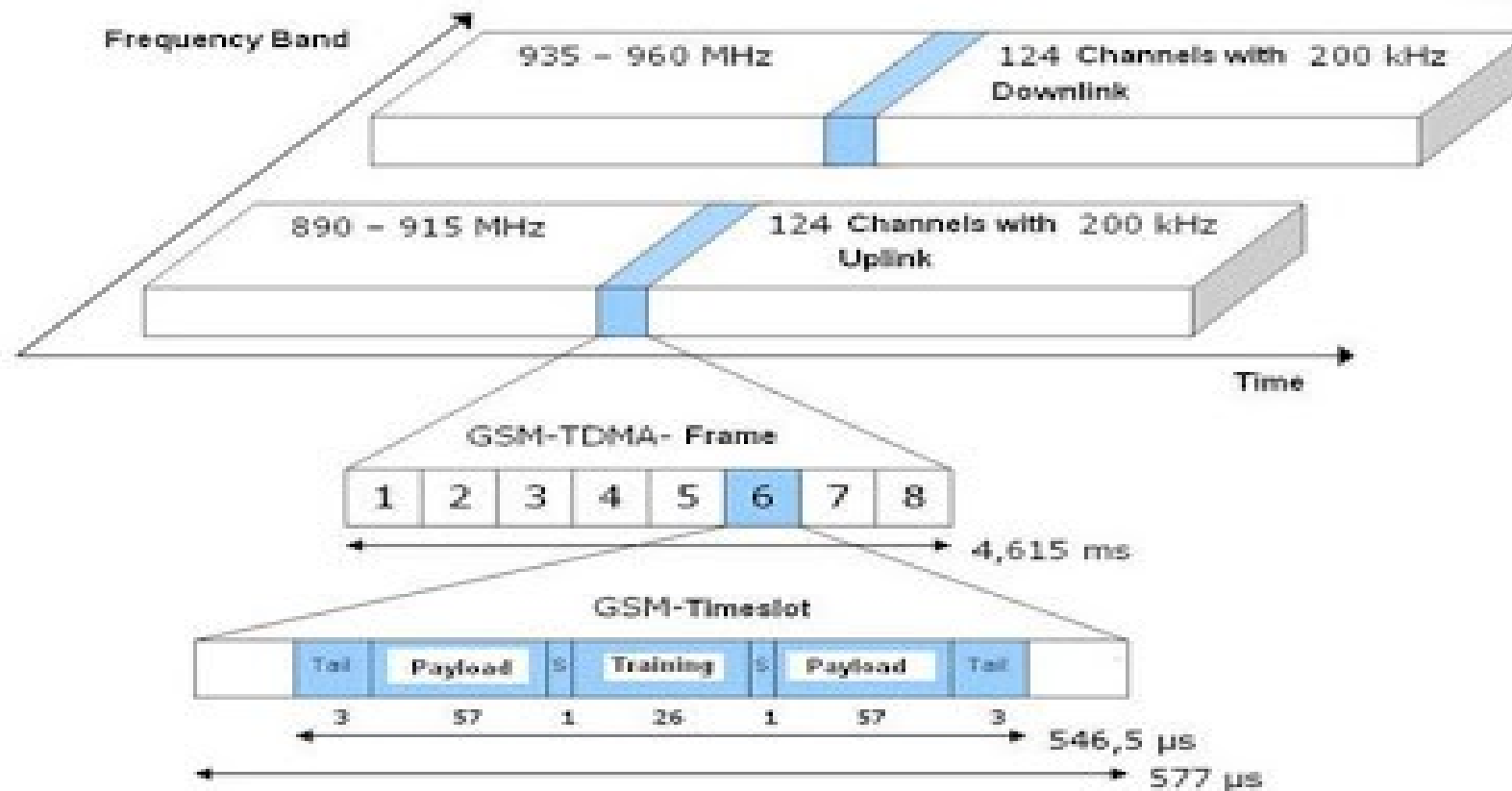


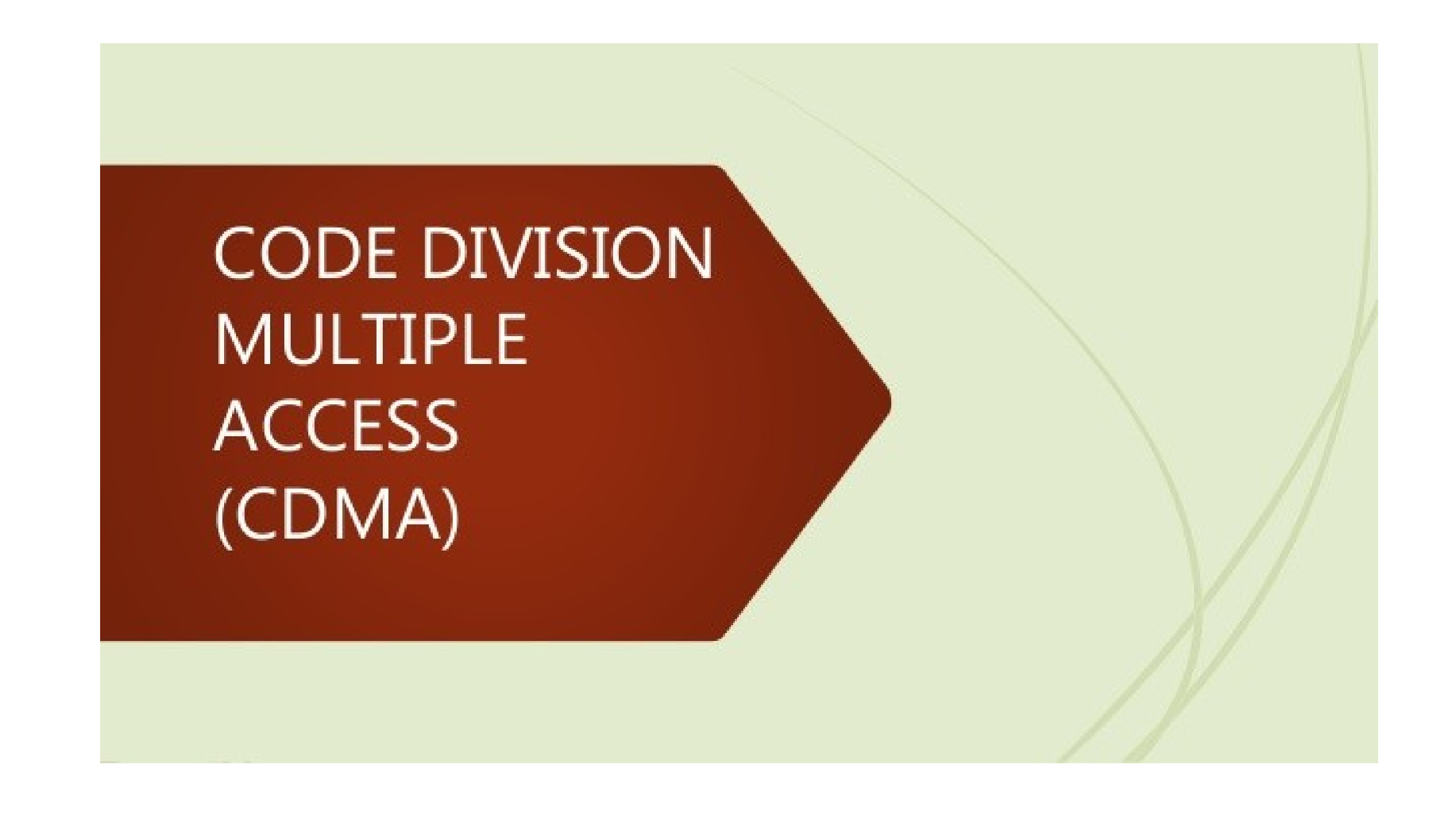
DECT: Digital Enhanced Cordless Telecommunications
TDD: Time Division Duplex

FDD/FDMA - general scheme, example GSM

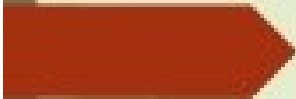


FDMA and TDMA in GSM





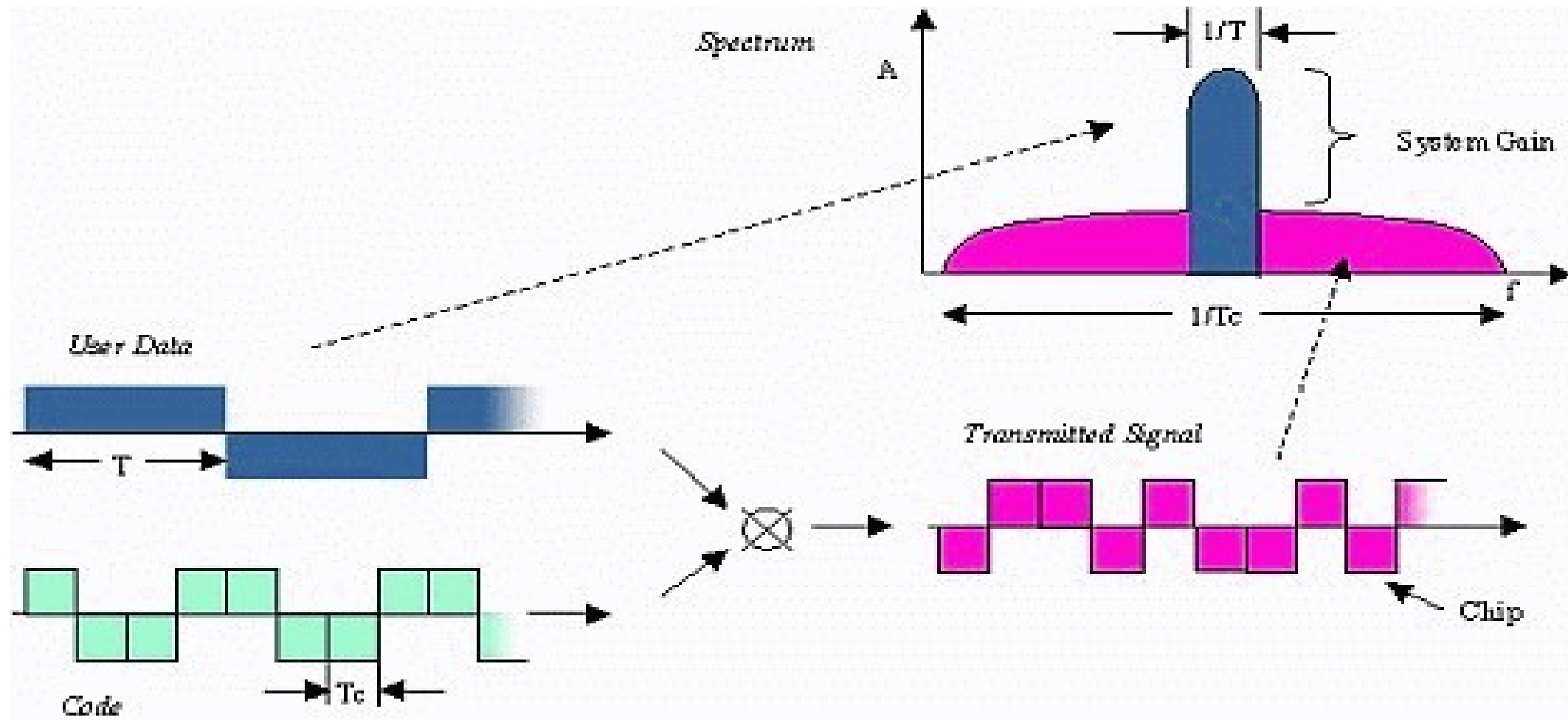
CODE DIVISION MULTIPLE ACCESS (CDMA)



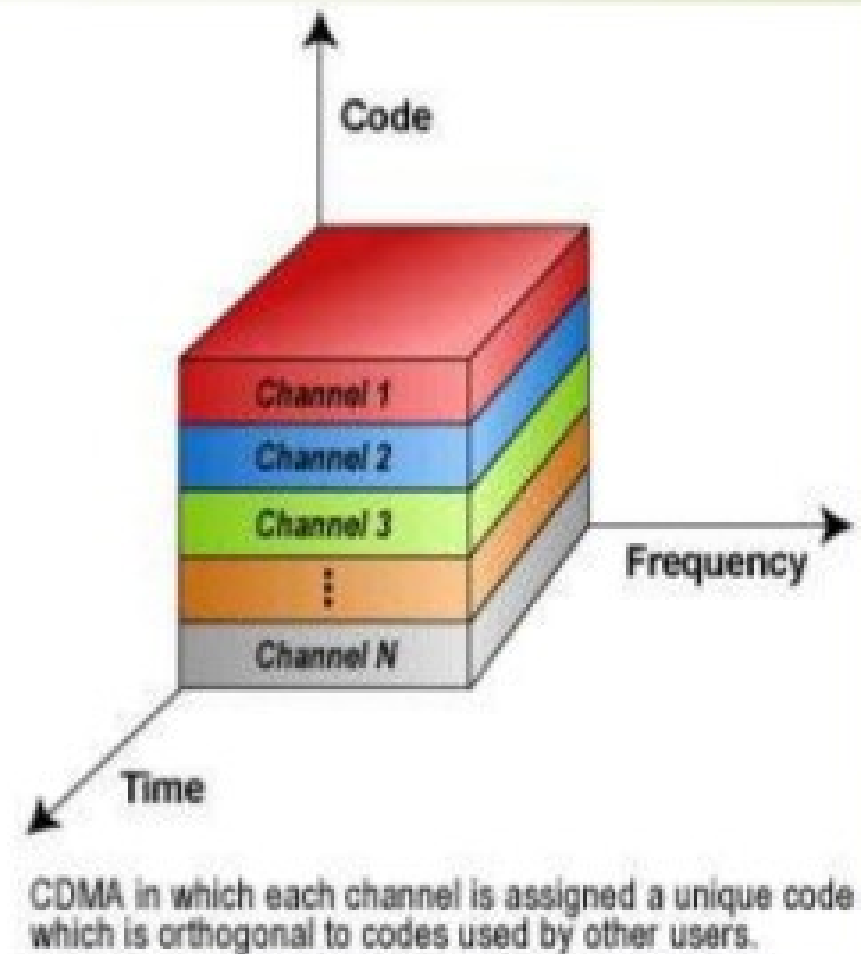
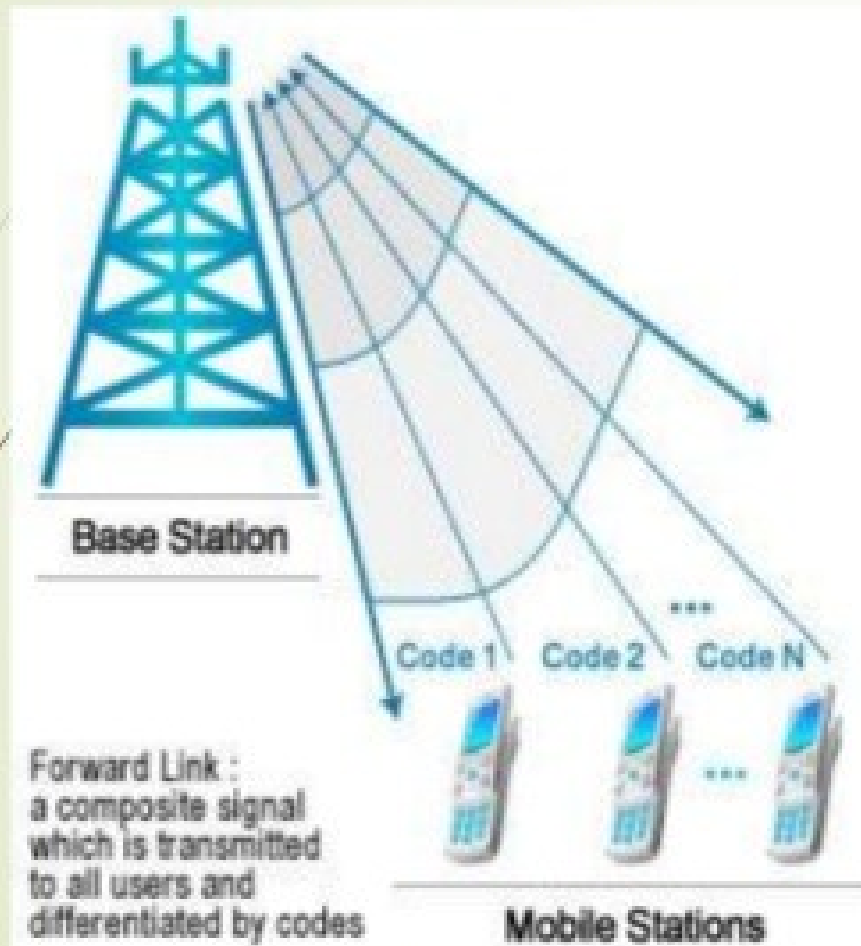
WHAT IS CDMA TECHNOLOGY?

- Code-division multiple access (CDMA) is a channel access method used by various radio communication technologies.
- CDMA is an example of multiple access, where several transmitters can send information simultaneously over a single communication channel.
- This allows several users to share a band of frequencies.

DSSS



CONT...



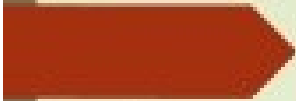
CDMA

- CDMA (Code Division Multiple Access)
 - There is no restriction on time and frequency in this scheme.
 - Parallel communication without collision and whole bandwidth can be used
 - Users are separated by code not by time slot and frequency slot
 - All terminals send on the same frequency probably at the same time and can use the whole bandwidth of the transmission channel
 - Each sender has a unique random number, the sender XORs the signal with this random number
 - The receiver can “tune” into this signal if it knows the pseudo random number, tuning is done via a correlation function



ADVANTAGES

- Improvement in capacity and security
- Improvement in hand over/ hand off
- Use of Wide Bandwidth
- It has more number of users can share the same bandwidth.
- It is well-matched with other cellular technologies.



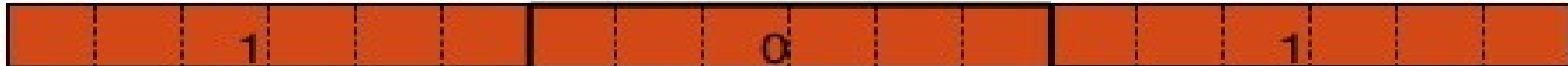
DISADVANTAGES

- The system is more complicated.
- Guard band and guard time both are required to be provided.
- As the number of users increases, the overall quality of services decrease

CDMA in theory

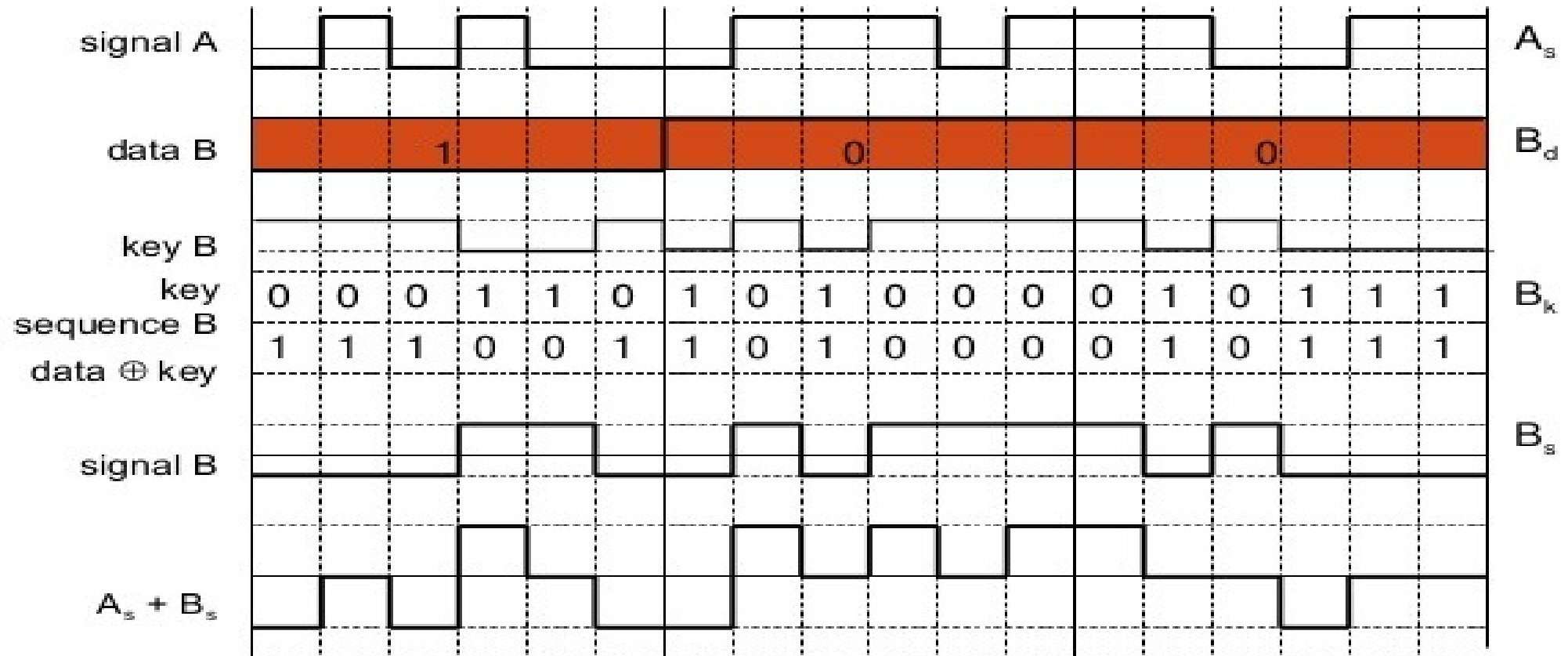
- Sender A
 - sends $A_d = 1$, key $A_k = 010011$ (assign: "0" = -1, "1" = +1)
 - sending signal $A_s = A_d * A_k = (-1, +1, -1, -1, +1, +1)$
- Sender B
 - sends $B_d = 0$, key $B_k = 110101$ (assign: "0" = -1, "1" = +1)
 - sending signal $B_s = B_d * B_k = (-1, -1, +1, -1, +1, -1)$
- Both signals superimpose in space
 - interference neglected (noise etc.)
 - $A_s + B_s = (-2, 0, 0, -2, +2, 0)$
- Receiver wants to receive signal from sender A
 - apply key A_k bitwise (inner product)
 - $A_s = (-2, 0, 0, -2, +2, 0) \bullet A_k = 2 + 0 + 0 + 2 + 2 + 0 = 6$
 - result greater than 0, therefore, original bit was "1"
 - receiving B
 - $B_s = (-2, 0, 0, -2, +2, 0) \bullet B_k = -2 + 0 + 0 - 2 - 2 + 0 = -6$, i.e. "0"

CDMA on signal level I

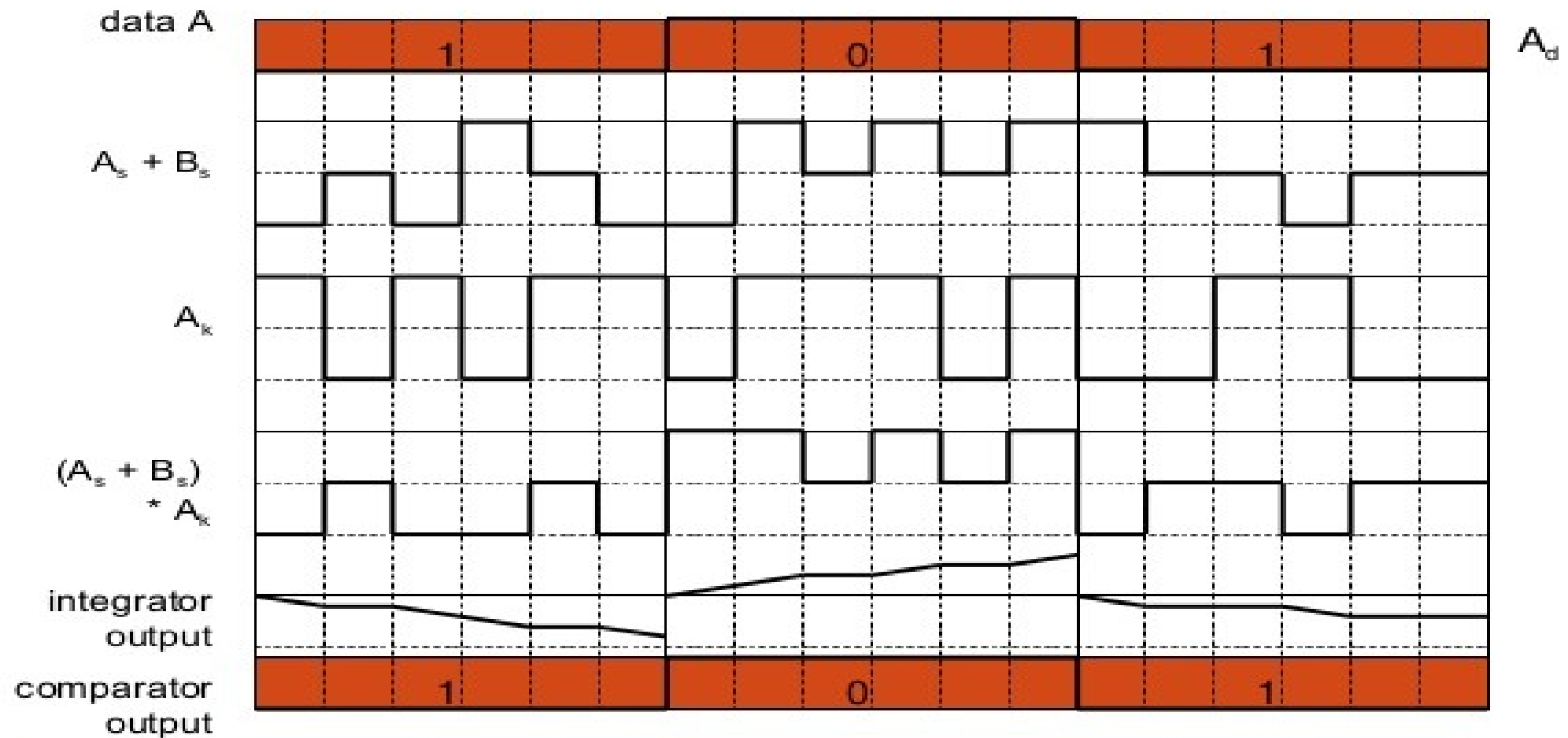
data A																	A_d		
key A																			
key sequence A	0	1	0	1	0	0	1	0	0	0	1	0	1	1	0	0	1	1	A_k
data $\oplus$ key	1	0	1	0	1	1	1	0	0	0	1	0	0	0	1	1	0	0	
signal A																	A_s		

Real systems use much longer keys resulting in a larger distance between single code words in code space.

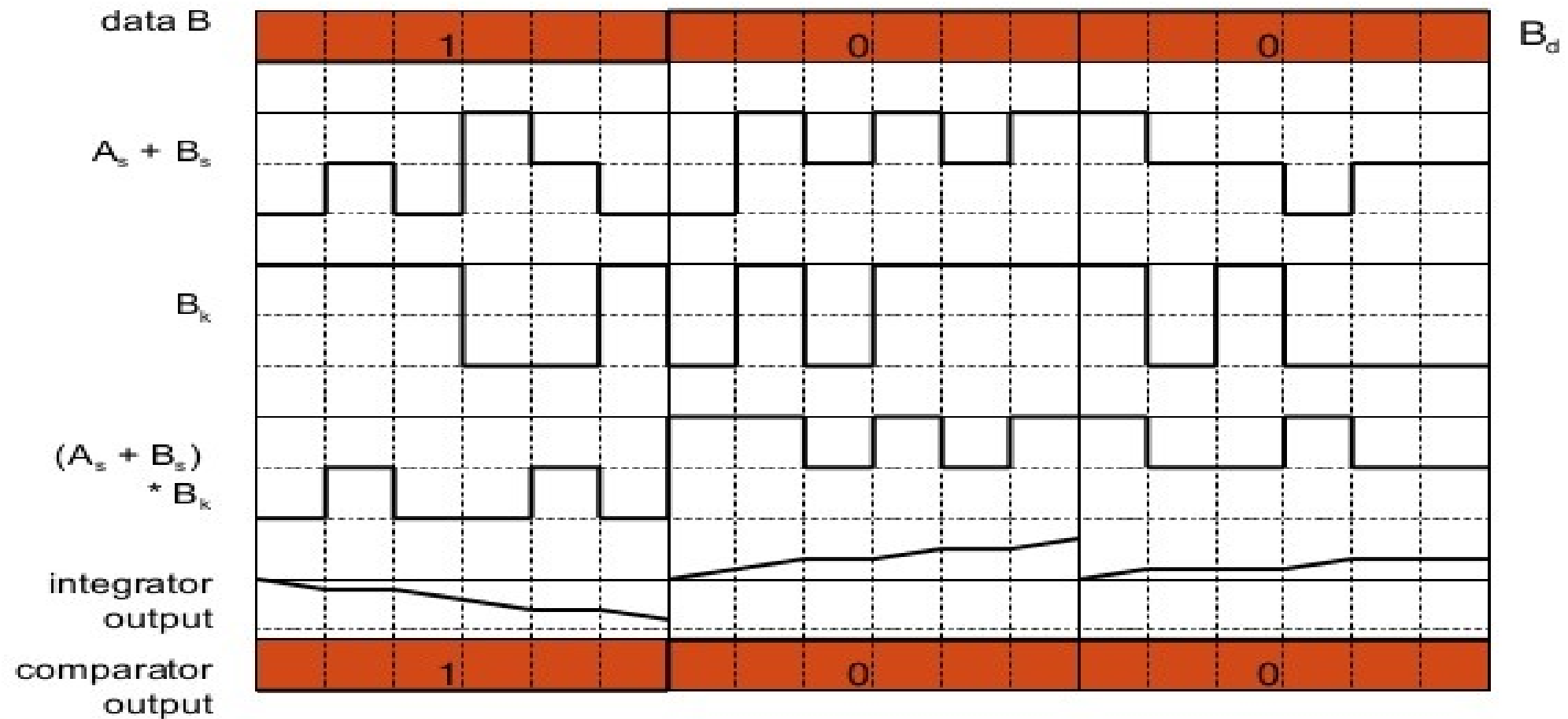
CDMA on signal level II



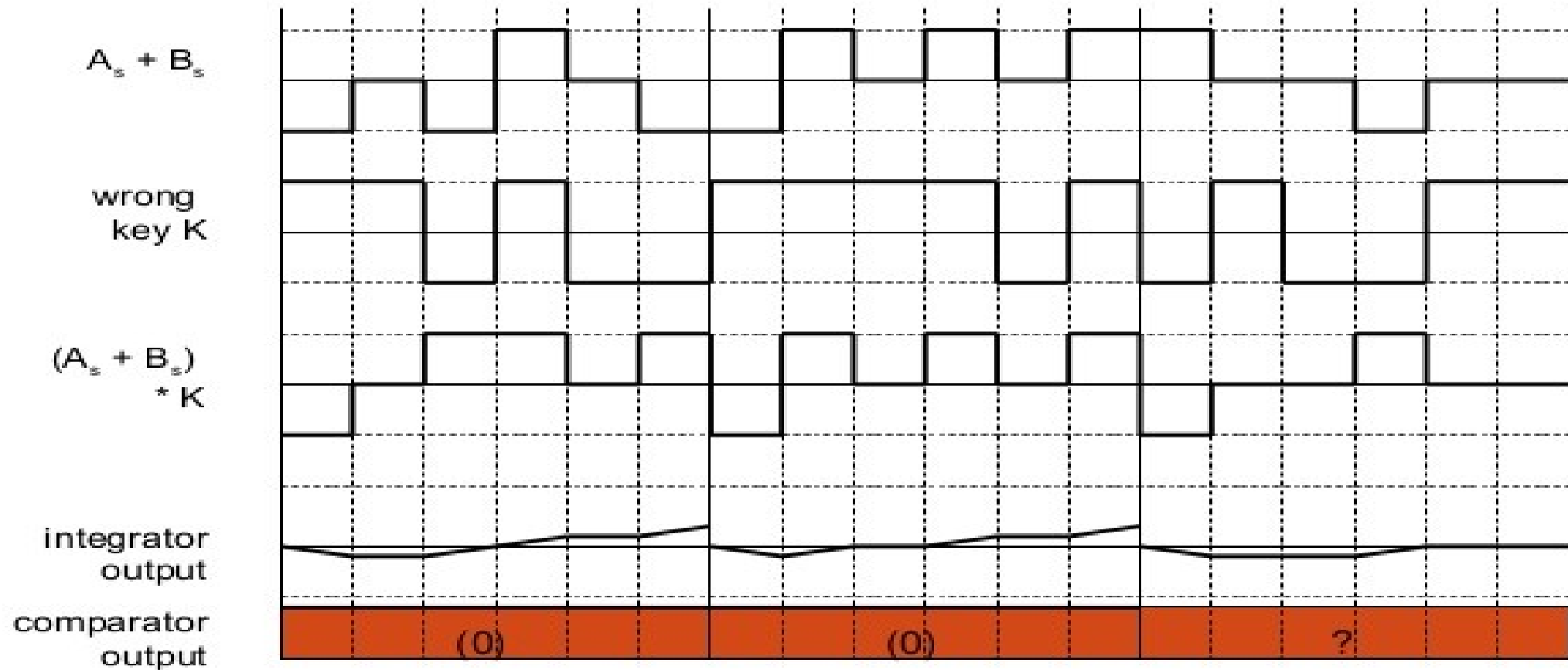
CDMA on signal level III



CDMA on signal level IV



CDMA on signal level V

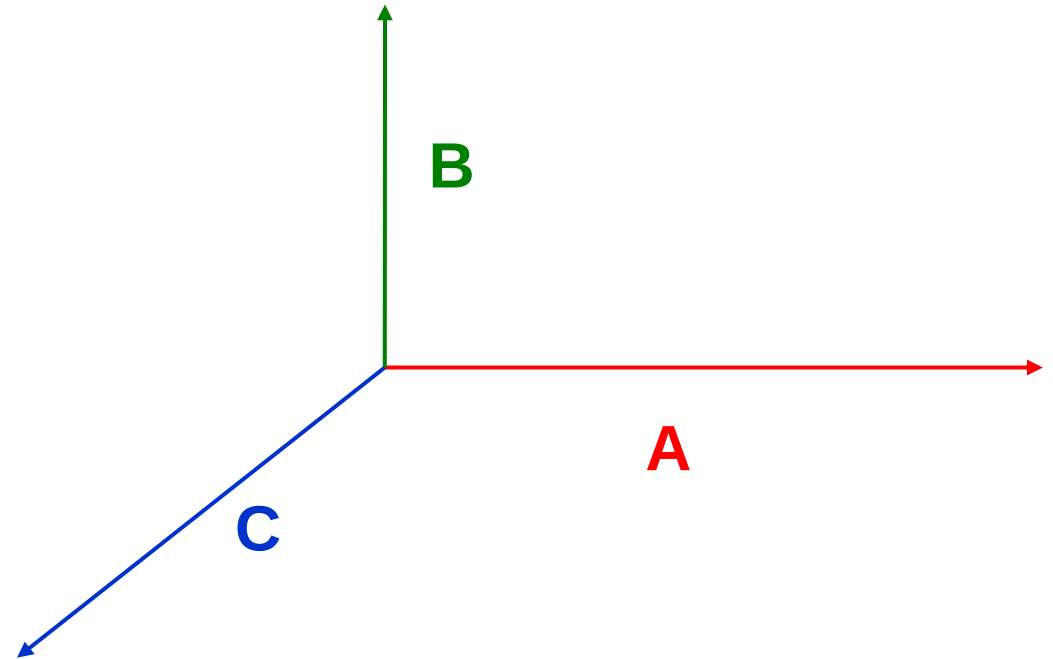


Orthogonal Frequency Division Multiple Access (OFDMA)



Orthogonality Principle

- **Vector space**
 - **A**, **B** and **C** vectors in space are orthogonal to each other
 - **$A \cdot B = B \cdot C = C \cdot A = 0$**
 - **$(A+B+C) \cdot A = (\text{mod } A)^2$**
 - **$(A+B+C) \cdot B = (\text{mod } B)^2$**
 - **$(A+B+C) \cdot C = (\text{mod } C)^2$**



Orthogonality Principle cont..

- Real Function space

$$f_1(t) = A \sin(\omega t)$$

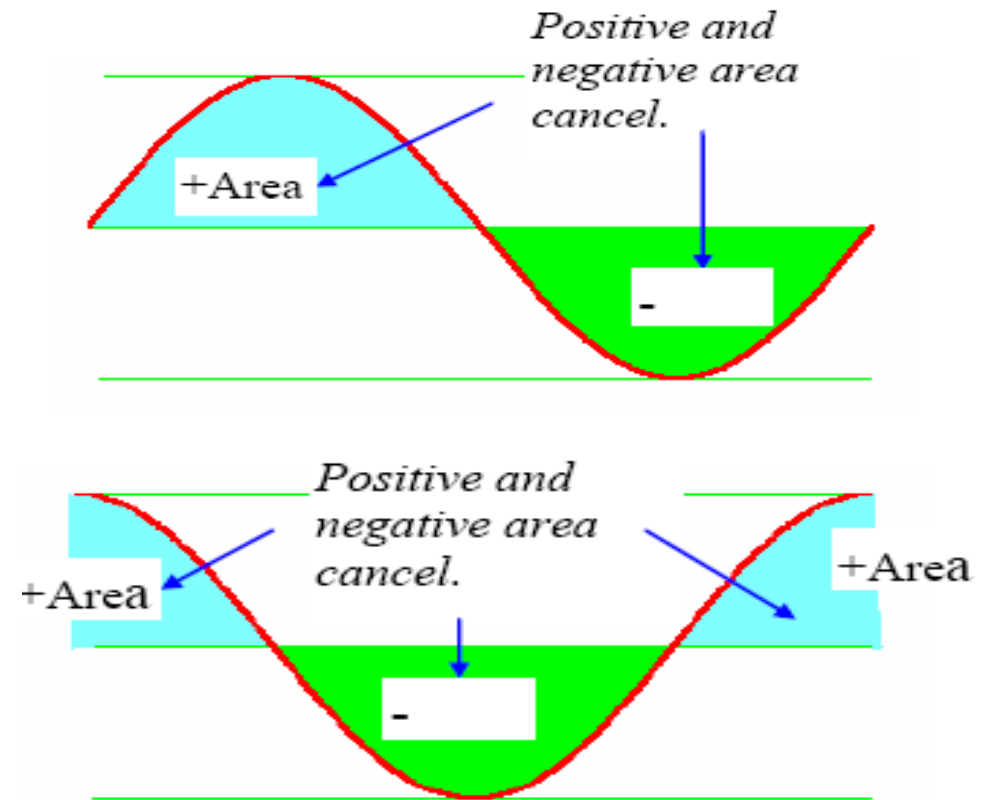
$$f_2(t) = B \cos(\omega t)$$

$$\int f_1(t) f_2(t) dt = 0$$

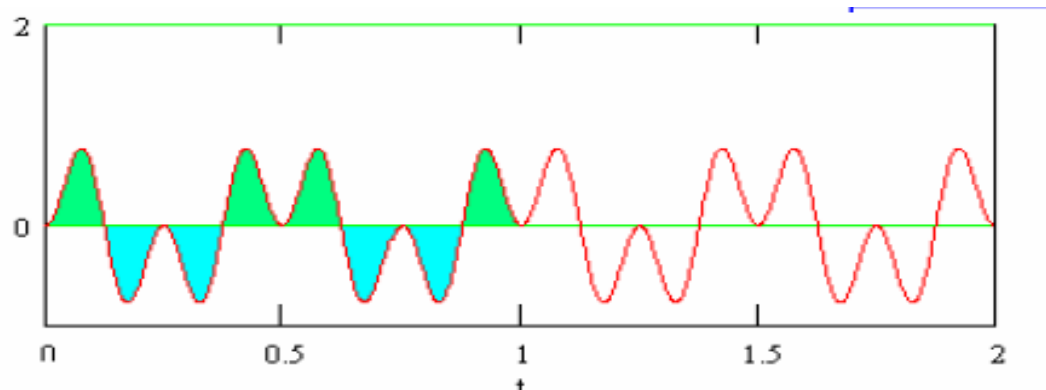
$$f_m(t) = M \sin(m\omega t)$$

$$f_n(t) = N \cos(n\omega t)$$

$$\int f_m(t) f_n(t) dt = 0$$



Orthogonality Principle cont..



$$f(t) = \sin(wt) \sin(2wt)$$

$$\forall m, n \in N$$

$$\int_0^T \sin(mwt) \sin(nwt) dt = 0 \text{ where } m \neq n$$

$$\int_0^T \sin(mwt) \cos(nwt) dt = 0$$

Here ***mw*** and ***nw*** are called
m-th and ***n-th*** harmonics of ***w***
respectively

Orthogonal Frequency Division Multiplexing(OFDM)

- It is a special kind of FDM
- The spacing between carriers are such that they are orthogonal to one another
- Therefore no need of guard band between carriers.

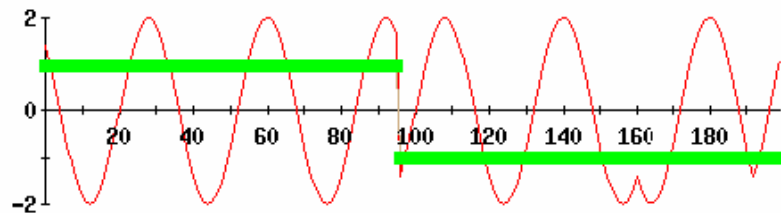
Example of OFDM

- Lets we have following information bits
 - 1, 1, -1, -1, 1, 1, 1, -1, 1, -1, -1, -1, -1, 1, -1, -1, ...
- Just converts the serials bits to parallel bits

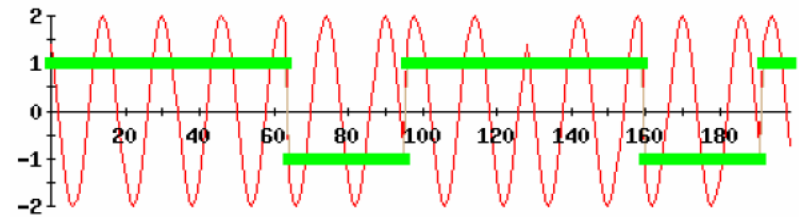
C1	C2	C3	C4
1	1	-1	-1
1	1	1	-1
1	-1	-1	-1
-1	1	-1	-1
-1	1	1	-1
-1	-1	1	1

Example of OFDM cont..

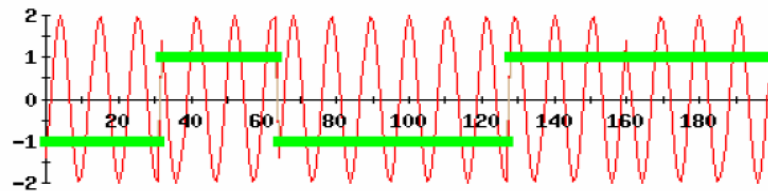
Modulate each column with corresponding sub-carrier using BPSK



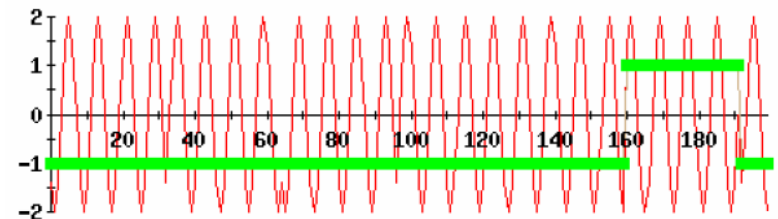
Modulated signal for C1



Modulated signal for C2



Modulated signal for C3



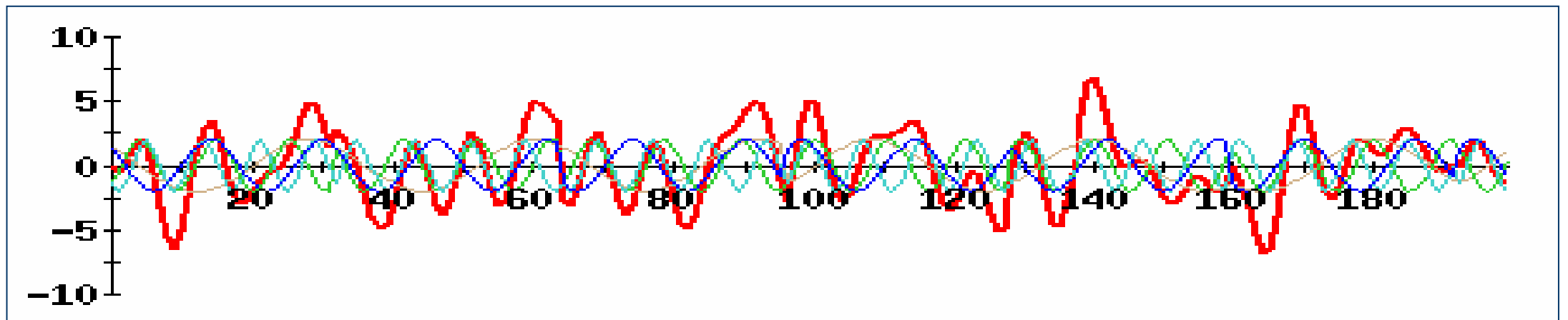
Modulated signal for C4

Example of OFDM cont..

- Final OFDM Signal = Sum of all signal

$$V(t) = \sum_{n=0}^{N-1} I_n(t) \sin(2\pi n t)$$

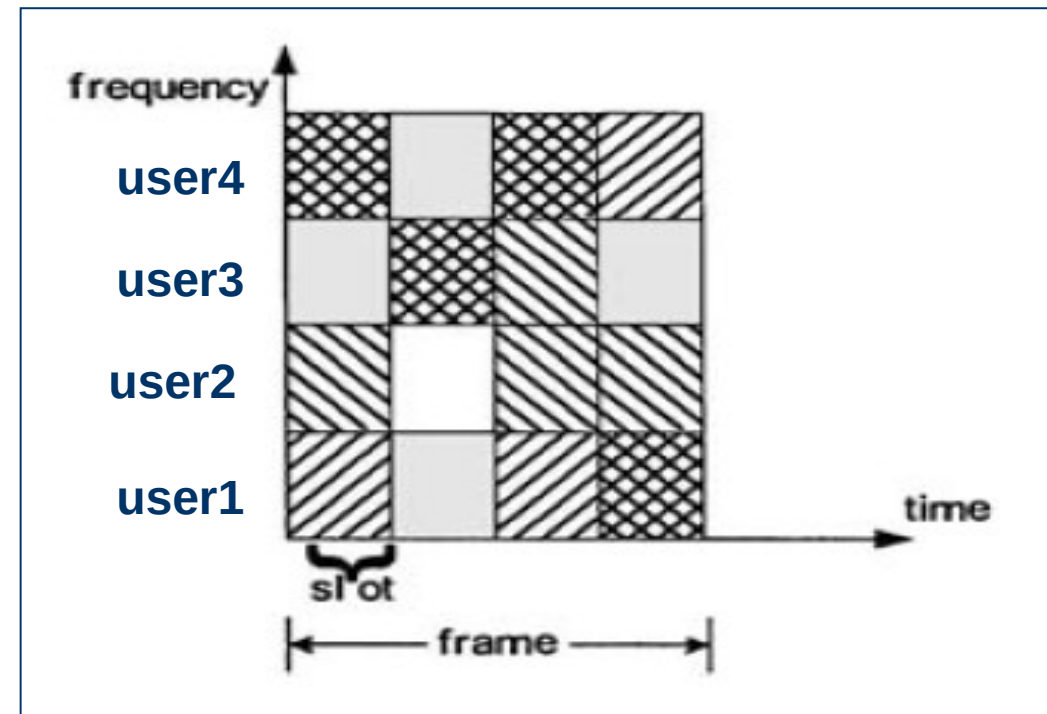
V(t) ———



Generated OFDM signal, V(t)

OFDM-FDMA (OFDMA)

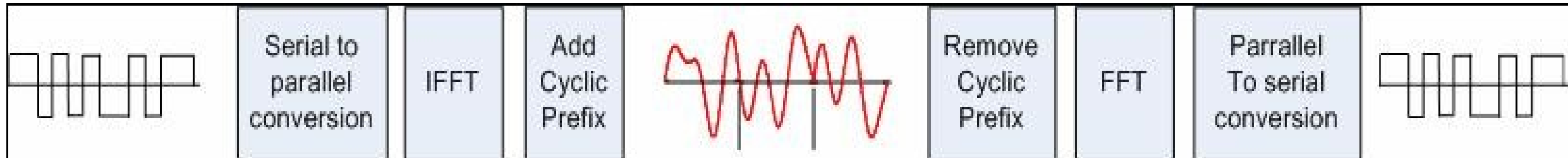
- Each terminal occupies a subset of sub-carriers
- Subset is called an OFDMA traffic channel
- Each traffic channel is assigned exclusively to one user at any time



OFDM-FDMA (OFDMA)

- *The IEEE 802.16e/ WiMax use OFDMA as Multiple access technique*
 - *Bandwidth options 1.25, 5, 10, or 20 MHz*
 - *Entire bandwidth divided into 128, 512, 1024 or 2048 sub carriers*
 - *20 MHz bandwidth with 2048 sub carriers has 9.8 KHz spacing between sub carriers*

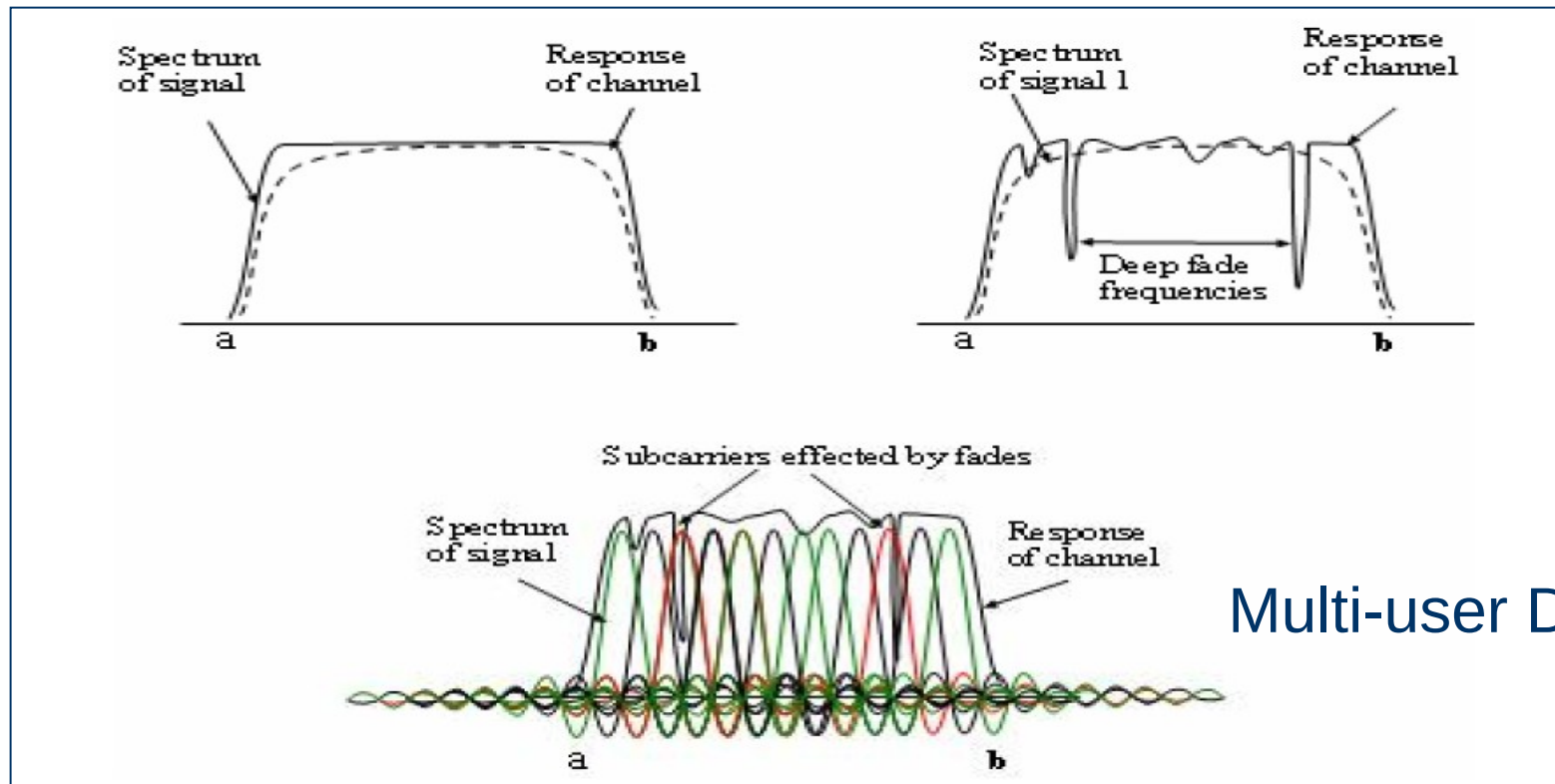
OFDM-FDMA (System View)



Advantages of OFDMA

- Multi-user Diversity
 - broadband signals experience frequency selective fading
 - OFDMA allows different users to transmit over different portions of the broadband spectrum (traffic channel)
 - Different users perceive different channel qualities, a deep faded channel for one user may still be favorable to others

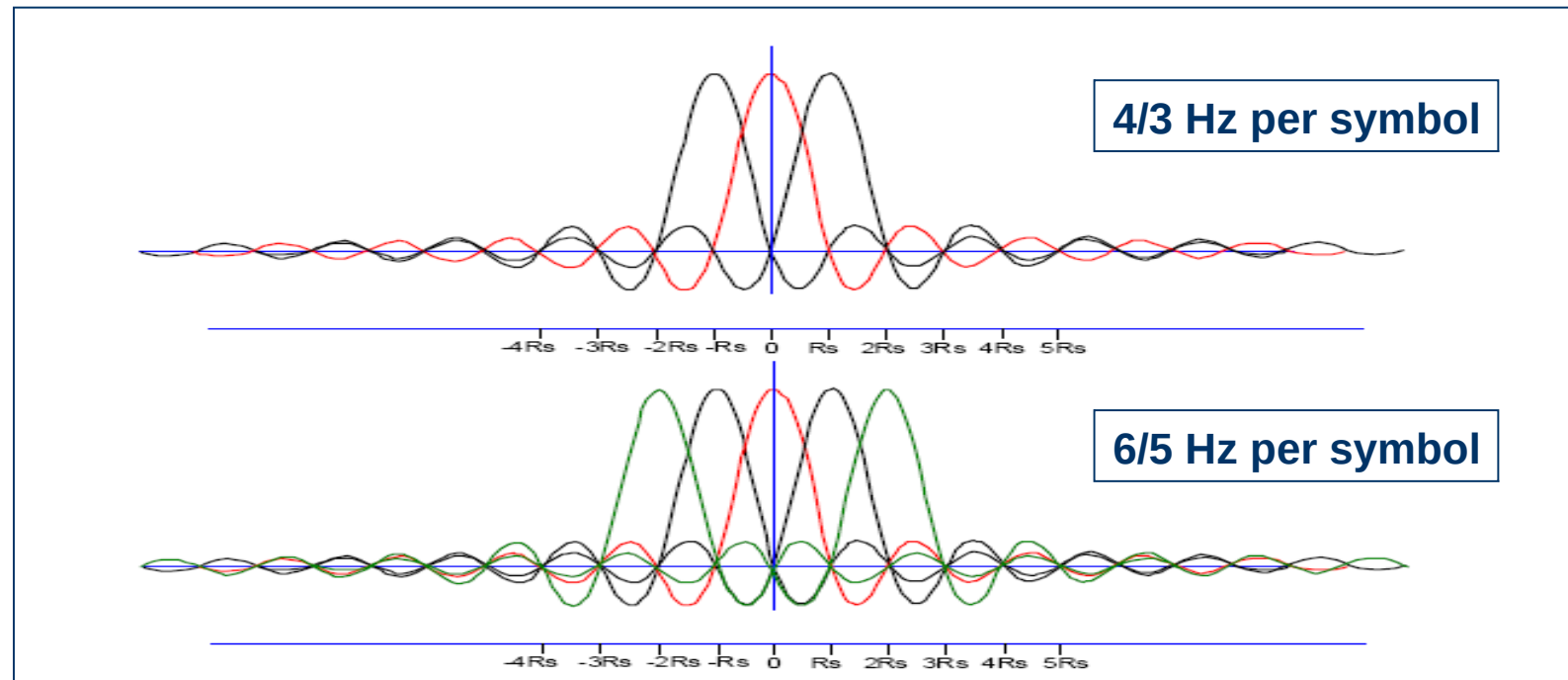
Advantages of OFDMA cont..



Multi-user Diversity

Advantages of OFDMA cont..

- Efficient use of Spectrum



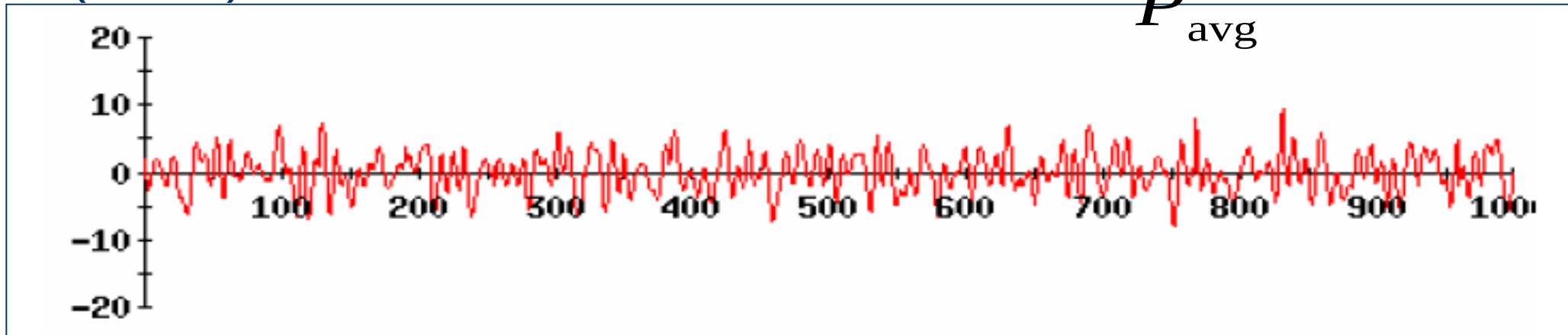
Advantages of OFDMA cont..

- Receiver Simplicity
 - It eliminates the intra-cell interference avoiding CDMA type of multi-user detection
 - Orthogonality of code destroyed by selective fading
 - Only FFT processor is required
- Bit Error Rate performance is better only in Fading environment

Disadvantages of OFDMA

- Peak to average power ratio (PAPR)

$$\text{PAPR} = \frac{|x(t)|^2}{P_{\text{avg}}}$$



The large amplitude variation increases in-band noise and increases the BER when the signal has to go through amplifier nonlinearities.

Disadvantages of OFDMA cont..

- Synchronization
 - Tight Synchronization between users are required for FFT in receiver
 - Pilot signals are used for synchronizations
- Co-channel interference
 - Dealing with this is more complex in OFDM than in CDMA
 - Dynamic channel allocation with advanced coordination among adjacent base stations